Summary:

GRAD-CAM

Grad-CAM was applied to the ResNet18 baseline model to answer one question: which frequency regions in the EEG spectrogram did the model focus on when deciding seizure or background? It does this by tracing the prediction backward through the network and highlighting the regions that influenced the decision most. This is different from raw signal strength; the model can ignore a loud frequency band and attend to a quiet one if it learned that band is more informative.

Model Performance

Overall accuracy: 79.2% *:* only slightly above the 79.3% majority-class floor
Seizure sensitivity: 56.0% : misses 44% of seizures, too low for clinical use
Background specificity: 85.3% : good at rejecting normal EEG

Key Finding: The Model Looks Beyond Raw Signal Strength

Raw EEG is delta-dominated in both classes (seizure 85%, background 97.5%), so signal power alone cannot tell them apart. Grad-CAM revealed the model learned something more useful. For seizure windows, attention was split across delta (51%), beta (26%), and gamma (13%). For background windows, it was split across delta (53%) and alpha (29%). Alpha suppression became the model's main background marker, while beta and gamma elevation became its main seizure markers. Both of these patterns align with known clinical EEG knowledge, confirming the model learned genuine discriminative features rather than a simple amplitude shortcut.

Architectural Limitation

ResNet18's progressive downsampling collapses the feature map to roughly 2x2 pixels at the final layer, making Grad-CAM heatmaps extremely coarse across all classes. This is a documented architectural finding, not a code error. It shows that ResNet18's depth, which benefits natural image recognition, works against fine-grained interpretability for small EEG spectrograms.

Integrated Gradients

Integrated Gradients (IG) was applied to both the balanced model (Run 1, 83.5% accuracy) and the imbalanced fine-tuned model (Run 2, 84.3% accuracy) across 153,679 one-second EEG windows. For each window, IG identified which moment within that

second, early (0–83ms), Mid (83–166ms), or Late (166–250ms) contributed most positively to the model's classification decision.

In the imbalanced model, both seizure and background windows peaked in the Mid segment at 38.5% and 39.9% respectively, meaning the model relied on nearly identical temporal regions regardless of the true class. This suggests the model was not learning meaningfully different time-based features to distinguish seizure from background.

The balanced model told a slightly different story. Seizure windows still peaked in Mid at 38.6%, but background windows shifted to peak in Early at 38.5%. This represents a small but notable separation between the two classes, the balanced model appears to have learned that seizure-relevant activity tends to occur in the middle of the window, while background activity is more identifiable from the very beginning.

It is important not to overclaim this difference, as the gap between segments remains small across both models. However, the direction of the finding is meaningful: balanced training produced marginally more class-distinct attribution patterns, while fine-tuning on imbalanced data slightly reduced both accuracy and attribution separability.

Overall, IG revealed that both models rely primarily on distributed signal features across the full window rather than sharply localized temporal moments. This suggests the 1D CNN architecture may not be capturing the temporal evolution of seizure activity in a clinically structured way. A more temporally aware architecture, such as an LSTM or Transformer, may produce stronger and more interpretable temporal attribution in future work 😊